

The Impact of Algorithmic Exposure to Opposing Viewpoints on Social Media on Societal Violence and User Retention

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1. Introduction

Over the last decade, the use of social media has increased dramatically, changing many aspects of our social lives. One of these aspects is polarisation of opinions among individuals of societies that use social media in daily life. This polarisation is fueled by hate speech or "cyberhate", which Castaño-Pulgarín et al. (2021) define as offensive language motivated by ideologies that violate social norms, often amplified by the anonymity of social networks (Vasist et al., 2023; Arcila Calderón et al., 2024). As a result, online hate speech frequently precedes offline violence that might cause real and physical harm. For instance, the genocide against the Rohingya in Myanmar in 2017 started with the mass spread of harmful anti-Rohingya content on Facebook (Zaleznik, 2021). The primary goal of social media algorithms is to optimise profit, total time spent by users on the platform watching advertisements (Metzler & Garcia, 2024). Some academics, such as Masood & Ahmed (2025) attempted to tackle the issue of the role of social media algorithms in polarising societies, by proposing "Incidental Exposure" to opposing opinions in a political context (Masood & Ahmed, 2025). They found that the increase of the incidental exposure is correlated with high political tolerance. However, Bail et al. (2018) argue that the exposure to a different viewpoint can lead to increase political polarisation and create a backfire effect, fostering users deeper in their radicalization. Addressing these two opposing findings and their effect on violence is empirically difficult, as testing these theories on actual populations could cause real world harm. Consequently, this study utilizes NetLogo simulation to safely simulate these dynamics. We specifically aim to answer the question of ***"To what extent does the change in the rate of algorithmic exposure to opposing viewpoints on social media affect the overall population violence and users leaving the platform?"***

2. Model Description and Choices

2.1. The Virtual Environment and User Attributes

In order to simulate the dynamics of social media algorithmic feed, we model the grid of the simulation as an abstract social space rather than a physical location. This utilises the Social Distance Attachment (SDA) framework proposed by Talaga & Nowak (2020), where interaction is taking place in this abstract space based on the closeness of the agents (turtles). The agents represent the users of the social media platform (grid) on a single social topic (e.g. politics, religion, etc). The agents move one step forward per time step, or they move two steps if the algorithm hides a post, allowing that user to move further toward a potential post they can interact with. Additionally, agents move in a random direction to ensure that agents are exposed to different groups of the populations and that they are not stuck with a single cluster of users. A radius of 5 implies that the content they are interacting with is overlapping, such as the agents are seeing the same comment section. The radius 5 represents the users' screens that can only fit the posts of the neighbour agents within the 5 radius. This design enables the simulation of scrolling or refreshing of users' feeds that change dynamically. That is consistent with Wong et al. (2006) who argue that in spatial models, the probability of connection decays with distance. Each user has a set of state variables that influence their behaviour over time (see diagram 1):

- **Opinion:** The main variable, which is a float number within the interval $[0.0, 1.0]$ where 0.0 and 1.0 are two opposing extreme viewpoints, while 0.5 is a moderate opinion on the topic of the platform. If the opinion value increased from 0.0 or decreased from 1.0 to 0.5, the opinion moves away from being extreme to a more moderate stance. The opinion is initialised by the initial-distribution parameter, which is a drop-down 'Chooser' in the interface that sets up the distribution of the opinions among the population and it has three distributions. The first is Moderate Majority where the simulation starts with a stable society with opinions normally distributed with a mean of 0.5 and standard deviation of 0.3. The second distribution is the Polarized Majority that illustrates a society with the majority of its population being close to extreme opinion, such as a civil war. This Polarized Majority distribution assigns opinions on users using a bimodal distribution where 50% chance of mean = 0.2, and 50% chance of mean = 0.8 with 0.3 standard deviation for both. The last

type of initial-distribution is the random distribution where users are assigned a random floating point number from the interval $[0.0, 1.0]$.

- **Radicalization:** Is a float number within the interval $[0.0, 1.0]$ represents how entrenched a user is in their opinion and higher radicalization makes an agent more resistant to opposing views. The systematic review (Wolfwicz et al. 2021) argues that cognitive radicalization is opinions that can lead to terrorism actions and that less than 1% of those who hold radical beliefs, ideas, and opinions will ever engage in acts of radical violence. This leads to calculating radicalization from the opinion of the user in a way that only far extremists (e.g. 0.0 or 1.0) becomes very radical, whereas a slight deviation from moderate opinion will decrease the radicalization significantly.
- **Trust-in-Platform:** A value $[0.0, 1.0]$ representing user satisfaction. If this drops to 0.0, the agent leaves the simulation. It is initialised to 1 at the beginning of the simulation for all users.
- **Activity-Rate:** A probability score $[0, 100]$ determining how frequently an agent posts content. Radicalization serves as a key factor in setting up the activity-rate value. Weeks et al. (2017) found positive and a significant relationship between seeking out like-minded information and sharing political information on social media for stronger partisans of a political party.
- **Support-For-Violence:** Represents psychological factors affect violent behaviour and it takes a value from $[0.0, 1.0]$. Castaño-Pulgarín et al. (2021) outline psychological theories that attempt to explain the phenomenon of “cyberhate” and Wolfwicz et al. (2021) argues that the support for radical violence is one of the “Radical Attitudes” that are the cognitive outcome that precedes behavioral radicalization or the actual violence. We formulate this variable such that high support-for-violence is very rare among radical users.
- **Violence:** A threshold variable where $[0.0, 0.2)$ is peaceful, $[0.2, 0.5]$ is a normal anger, and values >0.5 indicate aggression. This variable is calculated as the support-for-violence variable plus a random float number within the $[0.0, 0.2)$ interval, representing influences (peer pressure, opportunity, substance use). We derived the formula from (Carlsson, 2021) which illustrates the influence of several factors (e.g. substance use, trauma and personal history) on violent extremism (VE) and violent actions.

- **Published?:** Is a boolean variable that starts as false and turns to true when a user publishes a post (Xiong & Liu, 2014).
- **Resistance-to-Persuasion:** Is a nonlinear parameter, whose value is normally distributed so that the majority of the users receive the value of 2 for resistance-to-persuasion and other agents obtain 1.5 or 3 (Xiong & Liu, 2014). It represents the agent's resistance to persuasion to calculate the probability of a user shifting their opinion to the opposing view point.

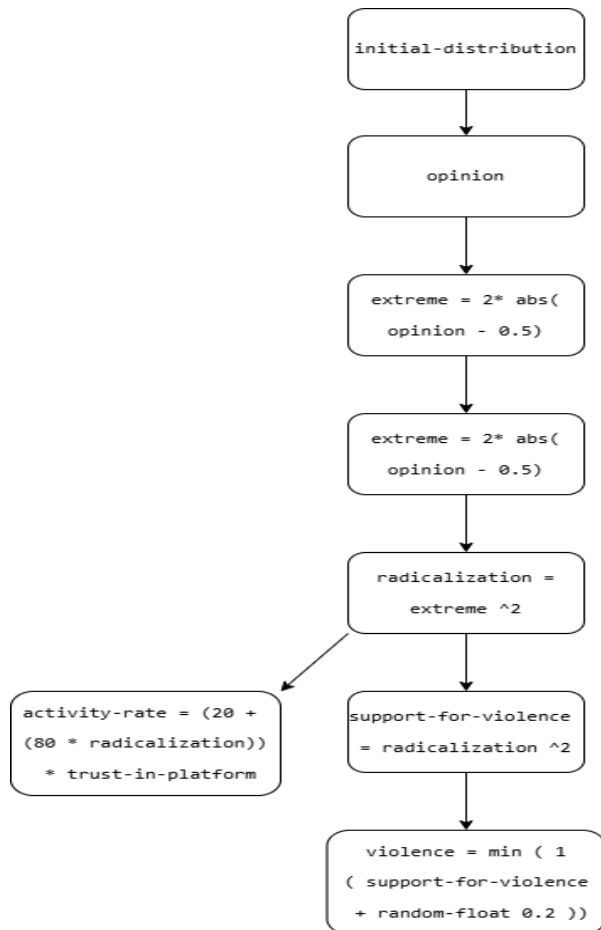


Diagram 1: the agent's variables formulas and flow

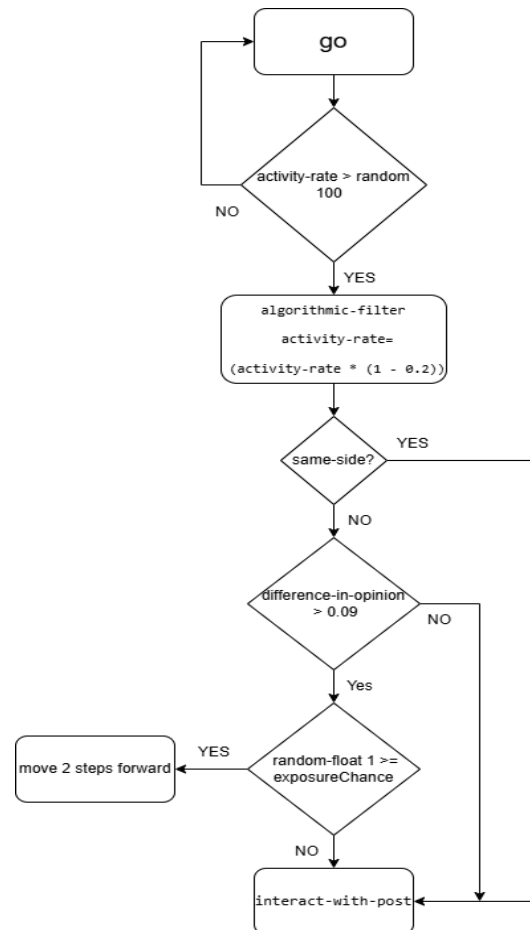


Diagram 2: the algorithmic filter for exposure chance parameter

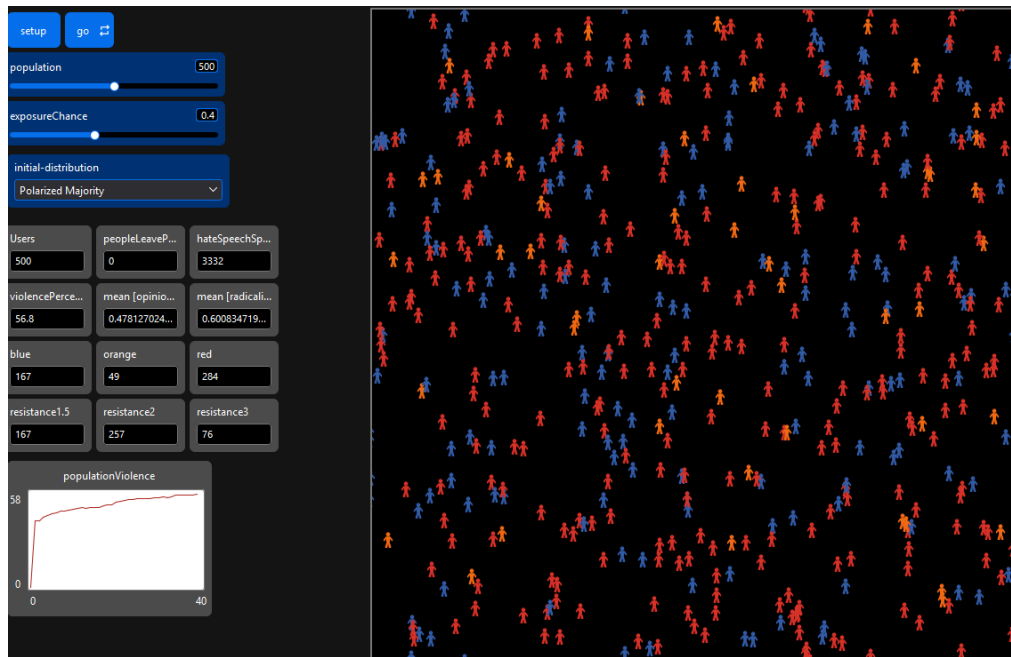


Image 1: overview of the interface of NetLogo, where we can see the parameters, monitors, and the plot of the violence level

2.2. Simulation Dynamics and Algorithmic Logic

Once all variables are set up, the simulation runs in discrete time steps or ticks. Initially, we check whether all users left the simulation or we reached the setup number of ticks. If this condition is met, the simulation stops and if not each agent checks some conditions before they are able to publish content (see Diagram 2). They check if their trust-in-platform is zero resulting in their removal from the simulation and if not, they go through an activity-rate check. If that check passes, there will be a decay factor of 0.2 applied, which was inspired by the simulation of Xiong & Liu (2014). They incorporate it to reflect the natural reduction in an agent's motivation or capacity to repeatedly take actions on social media.

Subsequently, the algorithmic-filter runs and it filters out the opposing viewpoints based on the exposureChance slider that ranges [0, 100]. A value of 0 means hide all posts that oppose the receiver's opinion, while 100 allows the receiver to view all posts including the ones with different opinions. Opinions are considered different if they lie on opposite sides of the moderate threshold (0.5) or if one is moderate (0.5) and the other is not. If the algorithmic-filter decides to show the receiver the post, the receiver runs the interact-with-post

function to simulate Echo Chamber Effect, Backfire Effect, or tolerance for moderates (Cinelli et al., 2021; Bail et al., 2018). Cinelli et al. (2021) define Echo chamber as environments where a user's political leaning, opinion, or belief gets reinforced due to repeated interactions with peers or sources that share similar tendencies. Consequently, for the Echo Chamber effect to be activated, both the viewer and the sender agents must have similar opinions that the absolute difference between the two opinions is less than 0.3. Another condition for the Echo Chamber is the probability of the receiver to change in their opinion. And we calculate that based on Xiong & Liu (2014) model where they used the formula $P = p^\alpha$ where p is the fraction of active (published? property) neighbors who have taken an external action opposing the agent's current opinion and α is the resistance-to-persuasion property. When the previous conditions are met, the Echo Chamber reinforces the receiver's opinion so they become 0.01 closer to the extreme opinion of their side and gain trust. Regarding the Backfire effect, it will push users away from the sender opinion and toward extremes and it increases the receiver trust after engagement via outrage (Bail et al., 2018). In order to trigger this effect, these conditions must be met. First, users must have different opinions, then the receiver's probability to change their mind must meet the condition of a random value drawn from $[0, 1] < \text{probability-change}$ (Cinelli et al., 2021). Finally, the receiver must have a radicalization value > 0.25 , meaning only extremists (Users with opinion < 0.25 or > 0.75) exhibit the backfire response. The last scenario in the function interact-with-post is when both users have different opinions and the receiver has a radicalization value < 0.25 or being more moderate. This pulls the receiver toward the different opinion using the Deffuant-Weisbuch model and decreases trust-in-platform by 0.005 after the discomfort of interacting with an opposing opinion (Deffuant et al., 2000).

3. *Experimental Setup*

3.1. *Hypotheses*

I. Hypothesis 1:

H1₀ (Null): Algorithmic exposure to opposing viewpoints has no significant effect on violence levels across all population distributions.

H1₁ (Alternative): Increased algorithmic exposure to opposing viewpoints significantly reduces violence levels across all population distributions.

II. Hypothesis 2:

H2₀ (Null): Algorithmic exposure to opposing viewpoints has no significant effect on the number of users leaving the platform.

H2₁ (Alternative): Increased algorithmic exposure to opposing viewpoints significantly increases the number of users leaving the platform across all population distributions.

III. Hypothesis 3:

H3₀ (Null): The effect of algorithmic exposure on violence is the same regardless of initial population distribution.

H3₁ (Alternative): The effect of algorithmic exposure on violence depends on the initial population distribution.

3.2. Scenarios and Parameters

To test these hypotheses, we initiate the population size to 500 and the grid dimension to 40 x 40 patches, which ensures that the mean number of neighbors within a radius of 5 is approximately 5 agents. That is necessary for calculating the p from the formula $P = p^\alpha$ (Xiong & Liu, 2014). After that, we conducted a BehaviorSpace experiment with 6 levels of the slider exposureChance variable (0, 0.2, 0.4, 0.6, 0.8, 1) and 3 levels of the chooser initial-distribution ("Moderate Majority" "Polarized Majority" "Random"), resulting in 18 scenarios. To be able to draw a strong conclusion, we executed 50 repetitions for each scenario, resulting in a total of 900 runs. Additionally, we opt for 20000 ticks as a limit for each run because we tested each condition one time and we observed that the violenceLevel stabilizes after 20000 ticks. Our dependent variables are the violenceLevel, which is the percentage of the overall users with a violence value greater than 0.5, and peopleLeavePlatform, which represents the number of agents who left the simulation after losing trust in the platform.

3.3. Statistical Analysis

statistical analyses were performed using Python (using statsmodels and pandas libraries). To test our hypotheses, we conducted a Two-Way ANOVA to evaluate the main effects and interaction effects of exposureChance and initial-distribution on the dependent variables. Significant interactions were

further explored using Tukey's Honestly Significant Difference (HSD) post-hoc test with a family-wise error rate (FWER) of 0.05.

4. Results

Effect	Dependent Variable	df	F	p
Exposure Chance	Violence Level	5	1016.24	< .005
	Users Leaving	5	958.56	< .005
Initial Distribution	Violence Level	2	26.41	< .005
	Users Leaving	2	26.19	< .005
Interaction	Violence Level	10	140.80	< .005

Table 1: Summary of Two-Way ANOVA results for Violence Levels and number of users leaving the platform

4.1. Effect of Algorithmic Exposure on Violence (H1 & H3)

Overall, the statistical output illustrates that the change in exposure chance significantly affected the violence level ($F(5, 882) = 1016.24, p < .005$), supporting H1₁. However, this main effect was qualified by a significant interaction between exposure chance and the initial population distribution (

$F(10, 882) = 140.81, p < .005$). This confirms H3₁, indicating that the algorithm's impact varies depending on the society type.

In Figure 1, the nature of this interaction is distinct across groups. In the Moderate Majority (blue bars), increasing exposure from 0.0 to 0.6 significantly reduced violence levels, suggesting that incidental exposure promotes tolerance in stable societies. As for the Polarized Majority (orange bars), increasing exposure from 0 to 0.4 resulted in a dramatic decrease in violence (*Mean Difference* = -18.68, $p < .005$). Nevertheless, the violence level soared to approximately 100% when the exposure chance went from 0.4 to 0.6 (*Mean Difference* = +19.46, $p < .005$), consistent with the Backfire Effect.

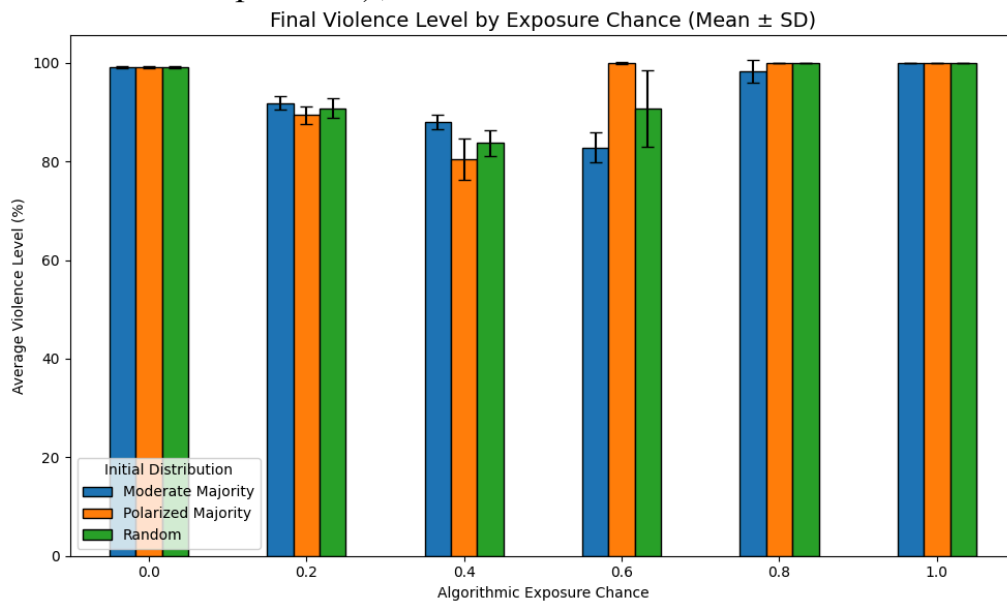


Figure 1

4.1. Impact on Users' Trust in the Platform (H2)

Regarding the sustainability of the platform, the results support H2₁. The ANOVA indicated a significant main effect of exposure chance on the number of users leaving the platform ($F(5, 892) = 958.56, p < .005$). Figure 2 illustrates this trend. Post-hoc analysis revealed that while low exposure levels (0.0 to 0.4) resulted in minimal users leaving the platform, high exposure levels (0.6 and above) caused a significant exodus. Specifically, shifting from 0.4 to 0.6 exposure caused a dramatic spike in users leaving (*Mean Difference* = 94.27, $p < .005$), regardless of the initial population distribution. This suggests a threshold effect where excessive exposure to opposing views alienates users, causing them to abandon the platform.

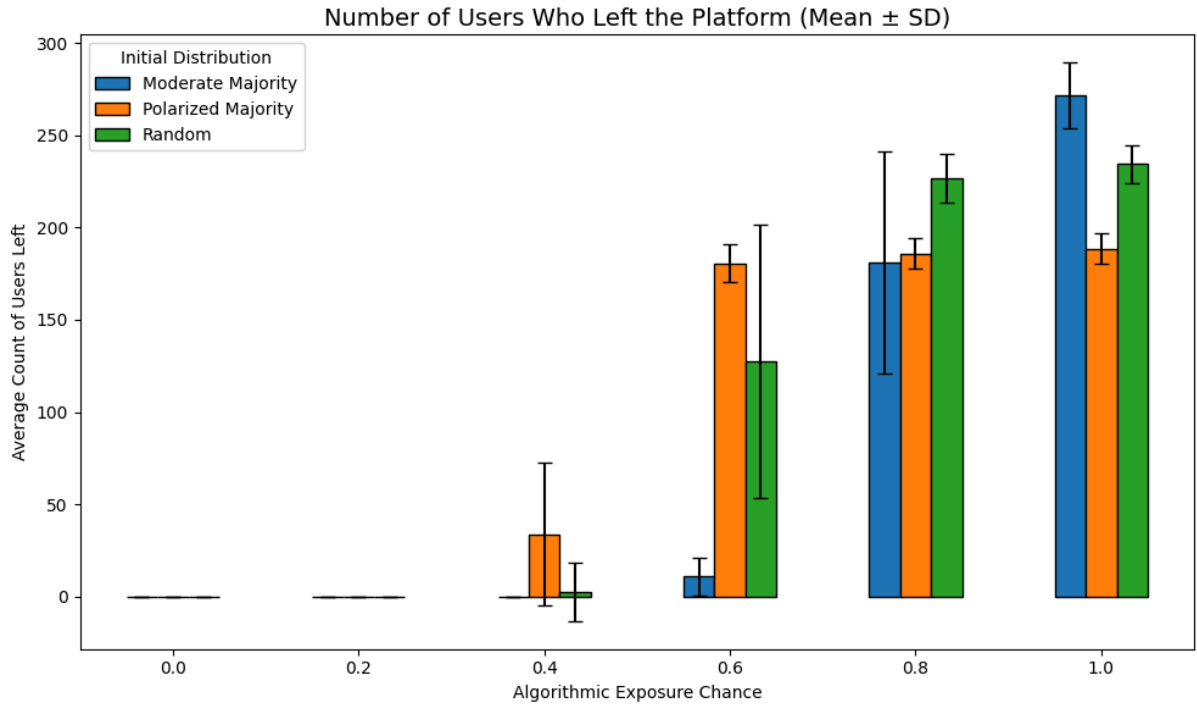


Figure 2

5. Discussion

The results resolve the theoretical conflict between Incidental Exposure (Masood & Ahmed, 2025) and the Backfire Effect (Bail et al., 2018) by identifying a critical tipping point at 0.4 (40%) exposure. Our data suggests that 0.4 represents a threshold and up to this point, violence decreases significantly across all population distributions. However, the effects diverge drastically in the Moderate Majority, increasing exposure to 0.6 further reduced violence (*Mean Difference* = -5.23), supporting the Incidental Exposure. Conversely, in the Polarized Majority, crossing the 0.4 threshold triggered a massive Backfire Effect, spiking violence levels by 19.46 points ($p < .005$). This suggests that while high exposure benefits stable societies, it is dangerous for polarised ones. Therefore, to minimize harm in an unpredictable world, algorithms should arguably cap exposure to opposing views at roughly 40%. Regarding H2, the results indicate a clear tolerance threshold. When exposure to opposing views exceeded 60% (*exposureChance* > 0.6), the number of users abandoning the platform soared. Interestingly, the high standard deviation in the "Random"

distribution at 0.6 exposure (STD = 73.99) suggests that in unstructured societies, the collapse of the user base is highly unpredictable.

5.1. Limitations

While these results are significant, the model has specific limitations. First, regarding the statistical analysis of users leaving the platform (H2), the data showed high heterogeneity of variance. As seen in the results, low exposure settings had an STD of 0.00, while high exposure settings reached an STD of 73.99. While the F-value was sufficiently large to suggest a real effect, this violation of ANOVA assumptions implies that the results might be less precise than reported. Secondly, the model initializes all agents with a trust-in-platform value of 1.0 (Maximum). In reality, users join social networks with varying levels of skepticism. This design choice likely delayed the leaving effect, as agents had a large initial trust. A more realistic distribution might show users leaving much earlier.

6. Conclusion

This study aimed to determine the extent to which algorithmic exposure to opposing viewpoints affects population violence and quitting the platform. Based on the simulation results, we conclude that algorithmic exposure is beneficial only up to critical thresholds of 40% (0.4) for polarised population and 60% (0.6) moderate population. Our findings indicate that up to this threshold, exposure successfully reduces violence across all society types. However, exceeding these limits yields diminishing returns for moderate populations while triggering a catastrophic "Backfire Effect" in polarized ones, where violence spiked by over 19 points. Furthermore, exposure levels above 0.6 caused a significant portion of users to abandon the platform entirely. Consequently, rather than maximizing diversity at all costs, social media algorithms should implement 40% of algorithmic exposure to balance violence reduction with users' trust and societal stability. Future research should address the limitations of the current spatial interaction model. Currently, agents interact based on proximity within an abstract grid. A more realistic iteration should replace this with an explicit link based using a Scale-Free Network (Barabási & Albert, 1999). Additionally, future models

should introduce a dynamic trust variable that fluctuates based on peer influence, rather than only decaying based on negative content exposure.

Declaration on the Use of Generative AI

I declare that I used Gemini as a writing assistant to improve grammatical accuracy, spelling, and academic sentence structure after I wrote the report draft. I personally conducted the simulation, wrote the code, performed the statistical analysis in Python, and made all interpretations of the data.

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